

$\Rightarrow$ Convolution Theorem:- it is use for finding the inverse of the Laplace.

General form:

$$\mathcal{L}^{-1} \{ \bar{P}(s) \} = f(s)$$

$$\mathcal{L}^{-1} \{ \bar{g}(s) \} = g(s) \quad \text{then}$$

$$\mathcal{L}^{-1} \{ \bar{P}(s) \cdot \bar{g}(s) \} = \int_0^t f(u) \cdot g(t-u) du.$$

$\Rightarrow$ Now

$$\text{ex!} \quad \mathcal{L}^{-1} \left\{ \frac{s}{(s^2+a^2)^2} \right\} = \mathcal{L}^{-1} \left\{ \frac{s}{(s^2+a^2)} \cdot \frac{1}{(s^2+a^2)} \right\}$$

$$\bar{f}(s) = \mathcal{L}^{-1} \left\{ \frac{s}{s^2+a^2} \right\} = \cos at$$

$$\bar{g}(s) = \mathcal{L}^{-1} \left\{ \frac{1}{s^2+a^2} \right\} = \frac{1}{a} \sin at$$

Now By using conv theorem.

$$\int_0^t \cos au \cdot \frac{1}{a} \sin a(t-u)$$

$$\frac{1}{a} \int_0^t \cos au \sin a(t-u)$$

$$\frac{1}{a} \times \frac{1}{2} \int_0^t 2 \cos au \cdot \sin a(t-u)$$

$$\frac{1}{2a} \int_0^t 2 \cos au \cdot \sin a(t-u)$$

~~Prove~~ $2\cos(A+B)$

Formula: $2\cos A \sin B = \sin(A+B) - \sin(A-B)$

Now

$$\frac{1}{2a} = \int_0^t 2\sin\left[\frac{1}{2}au + \frac{1}{2}a(t-u)\right] - \sin\left[\frac{1}{2}au - \frac{1}{2}a(t-u)\right]$$

$$\frac{1}{2a} \int_0^t 2\sin a(u+t-u) - \sin a(u-t+u) du$$

$$\frac{1}{2a} \int_0^t 2\sin at - \sin(2u-t) du$$

$$\frac{1}{2a} \left[\sin at \right] \left[u \right]_0^t + \left[\frac{\cos a(2u-t)}{2a} \right]$$

$$\frac{1}{2a} \left[t \sin at + \frac{\cos at}{2a} - \frac{\cos at}{2a} \right] = t \left[\frac{\sin at}{2a} \right]$$

??

⇒ ex2 $\mathcal{L}^{-1} \left\{ \frac{6}{(s^2+9)^2} \right\}$ solve by using convol theorem:

$$\mathcal{L}^{-1} \left\{ \frac{6}{(s^2+9)^2} \right\} = \mathcal{L}^{-1} \left\{ \frac{1}{(s^2+9)} \cdot \frac{1}{(s^2+9)} \right\}$$

Let's

$$f(s) = \frac{6}{(s^2+9)} = \frac{1}{\cancel{6}} \boxed{\frac{\sin 3t}{3}}$$

$$g(s) = \frac{1}{s^2+9} = \boxed{\frac{\sin 3t}{3}}$$

$$\int_0^t \frac{\sin 3u}{3} \cdot g \frac{\sin 3(t-u)}{3}$$

$$\frac{1}{9} \int_0^t \sin 3u \sin 3(t-u)$$

$$\frac{2}{3} \int_0^t \frac{\sin 3u}{A} \cdot \frac{\sin 3(t-u)}{B}$$

$$\text{Formular } \sin A \cdot \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$\frac{2}{3} \int_0^t \frac{1}{2} [\cos(3u-3t+u) - \cos(3u+t-u)]$$

$$\frac{1}{3} \int_0^t [\cos(6u-3t) - \cos(-3t)]$$

$$\frac{1}{3} \int_0^t [\cos(6u-3t) - \cos(3t)] du$$

$$\frac{1}{3} \left[\frac{\sin(6t-3t)}{6} - t \cdot \cos 3t \right]$$

$$\frac{1}{3} \left[\frac{\sin 3t}{6} - t \cdot \cos 3t - \left(\frac{\sin(6t-3t)}{6} \right) \right]$$

$$\frac{1}{3} \left[\frac{\sin(6t-3t)}{6} - t \cdot \cos 3t - \frac{\sin(-3t)}{6} \right]$$

$$\frac{1}{3} \left[\frac{\sin 3t}{6} + \frac{\sin 3t}{6} - t \cdot \cos 3t \right]$$

$$\frac{1}{3} \left[\frac{\sin 3t + \sin 3t}{6} - t \cdot \cos 3t \right]$$

$$\frac{1}{3} \cdot \frac{1}{6} \left[2 \sin 3t - 6t \cdot \cos 3t \right]$$

$$\frac{1}{18} \left[2 \sin 3t - 6t \cdot \cos 3t \right]$$

$$\frac{1}{9} \left[\sin 3t - \frac{1}{2} t \cdot \cos 3t + \cos 3t \right]$$